Thanos Vassilakis

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Distinguished Engineer

MD, Head of AI and Architecture for Finance IT.

executive summary

Accomplished technology executive with over 20 years of experience driving transformation and innovation in the financial services sector. Demonstrated expertise in leading large-scale development initiatives, leveraging disruptive technologies, and fostering an entrepreneurial culture within established organizations. Proven record of enhancing income generation through strategic technology integration and operational efficiency. Adept at creating inclusive environments that empower teams and encourage collaboration to drive progress and maximize business impact. My career is characterized by a commitment to continuous learning, mentorship, and the pursuit of excellence in technology and its application in the finance industry.

current mandate

Technical ownership of GL migration from zOS/DB2 to Linux/Oracle and Sub-ledger modernization.
Drive modernization and AI adoption of RBC Finance IT.

what I've run at RBC

A multi national group of 166, 47% diversity, and with full-time teams in Canada, US, UK, Malaysia, Mexico, and consultants in India with a current budget of just over 30m.

Architecture for Operations, driving modernizing in reporting, monitoring, and the confirmation and settlement landscape, plus overseeing all development and vendor acquisitions in operations.
Access Control, this is a central access management services for 1k+ apps and services in Capital Markets
Enterprise Reconciliation Service, built in-house processing 300 realtime and 2k+ and daily reconciliations using ML and GPU matching.
Trade Expense Management.
Expense Reporting service for Capita Markets.
Global Market supervision an event driven services that is now using ML and hierarchical agents state service to model behavior and generate required reports by SEC, FINRA, UK and European regulators.
Market Access, Trade Limits Control. A Rule 15c3-5 compliant centralized ML automated trade limits for most all our order management systems.
ML Pricing Service. Uses by the Municipal Bond and Credit Derivative desks. Predicts a fair price for all unique 1.5m bonds in real-time and has beaten our competitor evaluation services every single trading day since it went live 4 years ago.
All Shared Operation Services, such as **UI Platforms**, Reporting Service, central queuing , Notification, FAS, **T+1 Reporting, + Trade Repository** for Capital Markets, Wealth Management and Cash Management
R&D for Technology - this included blockchain, crypto, ML model development, plus inner source and open source stewardship.

key achievements & deliverables

Strategic Cost Optimization & Vendor Consolidation

Enterprise Reconciliation Service (2024): Architected an in-house solution using ML and GPU matching to displace legacy vendor products, securing **$7M in annual savings**.
Trade Expense Services (2024): Deployed agentic invoice processing to modernize expense management, eliminating external vendor dependencies for **$11M/year savings**.
Global Market Supervision (2023): Engineered an event-driven, real-time supervision service that replaced third-party systems.
Post-Trade Modernization (2023): Built T+1 infrastructure and post-trade processing services, ensuring regulatory compliance.
Access Control Modernization (2023): Centralized application access control for over 1,000 apps.
Governance & Data (2023): Internalized **Reference Data Services** and **ML-based Trade Limits Governance**.

AI, Machine Learning & Quantitative Platforms

Pricing & Risk: Developed ML-based real-time pricing models for municipal bonds and credit derivatives, alongside real-time Position and PnL systems (2020).
ML Infrastructure: Delivered an ML Model execution and scoring service (2018) to operationalize data science initiatives.
Data Strategy: Established the Capital Markets Data Lake (2016) and a Single Data Interface SDK (2019) to standardize data access.

Core Infrastructure & Technology Transformation

Engineering Leadership: Built and scaled high-performance development teams across **Kuala Lumpur, Halifax, and Montreal**.
Language Adoption: Successfully introduced and standardized **Python** (2010) and **Elixir** (2017) within the enterprise technology stack.
Platform Engineering: Architected critical infrastructure including **Distributed Compute Service** (2018), **Function-as-a-Service (Lambda)** (2017), and **Central Logging** (2017).
Trading Cores: Built the **Real-time Trading Data Core** (Athena-style, 2018) and the **Cash Management Real-time Core** with virtual accounts.
Innovation: Developed a **Tokenization Platform** (2022) to advance digital asset capabilities.

Regulatory & Compliance

Risk & Reporting: Delivered infrastructure for **FRTB** (Fundamental Review of the Trading Book) (2018) and **Anti-Terrorism/Money Laundering** services (2018).
Exchange Migrations: Led the **LME trading platform** development and migrated commodity trades onto OpenLink.

previous work

Infinigon Group | Head of Development | 2012 - 2015
Created a cross-exchange custodial cryptocurrency trading platform.
Built a machine learning analytics platform utilizing real-time data and social media sentiment.

Credit Swiss | Contractor, Commodities | 2007-2010
Architected a risk system for complex commodities trades and integrated external pricing models for OpenLink
Built a central trade repository and market data services for commodities.

JPMorgan | Contractor, Energy Trading Systems | 2007
Contributed to the Athena platform, developing a code profiler and native FIX engine.
Developed a standardized trade feed framework and implemented gateways for CME and ICE to streamline energy trading operations.
Optimized network and computation performance for the counterparty risk system (Gauss).

Zingy | Head of Mobile Development | 2006 - 2007
Managed development and operations for 50+ mobile applications across 27 carriers and multiple platforms.
Led the development of innovative mobile products, including the top-selling MapQuest mobile application.

ICAP | Head of Front-End Development | 2004 - 2006
Established and led the in-house development team for fixed income trading systems.
Developed high-performance distributed trading services and mobile trading platforms.

NYSE/SEC/SIAC | Director, Specialist Trading Services| 1997 - 2004
Architected and implemented market surveillance systems, stock information servers, and regulatory reporting services.
Developed innovative Python-based trading services for specialists.

ERICSSON | Contractor, Telecom Switches | 1995 - 1997
Built a remote telecom switch management system.

bio

I came to Computer Science by accident. At eighteen, I had no interest in computers; my heart was set on being a hydrologist to solve water problems in developing countries. My technical interests were analog—trading and repairing vintage valve amplifiers. That changed during a summer job at the Electric Coin Company, a slot machine vendor. I expected to wield a soldering iron but found myself coding 6801 controllers. By the end of the summer, I was addicted to programming. I switched tracks, graduating from North London Polytechnic with a degree in Maths & Computer Science.

My early career was defined by eclectic, high-intensity technical challenges. I worked with Richard Forsyth on *Darwin*, one of the earliest Machine Learning frameworks, which we used to unsuccessfully predict horse races. I worked in Atari's PC games division, built RIP engines for printers, and developed a real-time OS for Ferranti's Lynx helicopter program. My freelance

work spanned industries: fuzzy logic control systems for Unilever, video routing for the BBC, and neural networks on Transputers for the UK Treasury.

In 1993, while working at Apple on the Newton program, I discovered the internet and built a WAIS-based patient record system for the UK National Health Service. A pivotal moment came in 1995 when I was commissioned to build the first internet casino for Caesars. My initial C++ codebase was tight but unwieldy at over 40,000 lines, and I struggled to complete it. On a whim, I bought Mark Lutz's Python book and rewrote the entire system over a weekend. The result was a superior, 4,000-line solution. That project made me a lifelong Python evangelist and led to the creation of an early cyber currency, WebCheck.

After stints in Europe—advising the European Parliament, building Erlang systems for Ericsson, and being poached by IBM Research in Zurich to code a mobile JVM—I returned to my hometown, New York, in 1997. I architected the NYSE's first internet-based system, which was likely the first financial production system written in Python. Since then, my focus has been on high-frequency trading, fixed income platforms, and risk systems for major institutions like ICAP, JP Morgan, Credit Suisse, and RBC Capital Markets. Between these corporate mandates, I continue to explore the bleeding edge through running a startup style group within the bank.

some key achievements & industry firsts

Europe's 1st Missing Children's Registry	Build a central online database of of missing children and was awarded a citation for the European Parliament.
Top earning US mobile app	MapQuest
Phonescript	1st phone programming language
eBay's first mobile app	Developed the initial mobile application for eBay
1st Python trading service	Built for SIAC/NYSE
1st C++ debugger & Numeric library	Coded standard libraries for Walter Bright's C++ distribution
1st Internet Casino	Caesars Casino
1st Chess app for iPhone	Reached to #23 in games in App Store
1st Web framework for Python (PSO)	Included in Debian Linux and used by European Parliament, NASA, SIAC, JP Morgan,
Byteeasy	Anonymous crypto exchange
BTC Terminal	Cross-exchange custodial crypto trading platform
JVM for Mobile	Coded the mobile phone version of the JVM for IBM
Author of Nimalana	Nim based Solana validator
Developer of Aegis	aZig based high-performance Bittensor miner specifically engineered to exploit Apple Silicon

table of domains & companies

Trading & Risk - Infinigon Group, RBC CM
Blockchain & Crypto - RBC CM, Infinigon Group, Solana, Marita Bank, Caesars Casino, Electric Coin Co.
Machine Learning & Big Data - RBC CM, Infinigon Group, Caesars Casino, Ericsson.
Mobile Development - Apple, RBC CM, IBM, ICAP, Ericsson, Zingy
Gaming & Robotics - Atari, Zingy, Caesars Casino, Unilever
High Performance Computing - JP Morgan,
Telecommunications - British Telecom, Ericsson, HP, Honeywell Bull, Multitone, Reuters
Cloud & Distributed Systems - RBC CM, Infinigon Group, Ericsson
Medical & Public Sector - British National Health, European Parliament, European Community, AAH Meditel
Gaming & Robotics - Atari, Zingy, Caesars Casino, Unilever
Military - Ferranti

stack of past work:

RBC CM, Infinigon Group, RBC CM, JP Morgan, Credit Suisse, JP Morgan, Zingy, ICAP, SIAC/NYSE, Ericsson/HP, West LB, European Community, Ceasars Casino, European Parliament, AAH Medical, Apple, Symantec (Zortech), Reuters International, Honeywell Bull, Drake, British Telecom, Ferrantti, Barclays, Multitone, Atari, Unilever, Atari, Electric Coin Company

enumeration of languages:

c, C++, erlang, elixir, fortran, go, java, javascript, python, solidity, zig, some rust, cobol

personal details

NATIONALITY: US & Greek (European)
SPOKEN LANGUAGES: English & French
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work references

Kabir Reike, MD Head of Data Architecture, Bank of America
Sid Belzberg, CFO, Infinigon Group - ex Captain of the Canadian Chess Team
Endri Mani, MD Head of GenAI , RBC CM
Yaroslav Chinskiy, MD, Head of Data Architecture, JP Morgan
Geraud Boyer, Director of Engineering, Datadog

personal references

Bruno Dupire, Head of Quantitative Research, Bloomberg
Hope Igelhart, Senior Psychoanalyst, Columbia University
Stavros Lambrinidis, European Union ambassador to the UN.